Live-Testbed Realization of QoE-Aware Deep Reinforcement Learning for O-RAN Network Slice Admission Control

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***Abstract*—**

TODO(MANOJ): Abstract. Suggested content sketch (2-3 bullets):

- **First live-testbed (not simulation) realization of DRL-based O-RAN slice admission control from papers #1/#2, on a real OAI gNB with the real E2 control-plane wire protocol – prior work was ns-3/ns-O-RAN simulation only.**
- **Headline quantitative result once Section V numbers are final (mean ± std across 3 seeds): SLA compliance / backlog comparison, static baseline vs. DQN/A2C under SLA and QoE-aware rewards.**
- **Testbed engineering contribution: empirical ratio-to-PRB calibration methodology, admission-as-ceiling-nudging realization, and a documented rig-fragility mitigation – reusable findings for anyone building a live O-RAN RL testbed.**

***Index Terms*—O-RAN, network slicing, deep reinforcement learning, admission control, quality of experience, live testbed**

I. Introduction

TODO(MANOJ): Introduction. Suggested content sketch:

- Motivation: O-RAN network slicing needs adaptive admission control; DRL is a natural fit but nearly all prior evaluation (including our own papers #1/#2) is simulation-only (ns-3/ns-O-RAN).
- Gap: no live-testbed validation of whether simulation-trained admission policies transfer to a real E2 control loop, a real MAC scheduler, and real per-slice traffic.
- Contribution list (map directly to Sections III-VI): (1) a working live O-RAN DRL admission-control testbed on real OAI hardware; (2) an empirical methodology for calibrating PRB ratio ceilings against real traffic (the ratio-to-PRB mapping is NOT 1:1 – a testbed-engineering finding in its own right); (3) a live comparison of a static O-RAN-native baseline vs. DQN/A2C under SLA (papers #1/#2 eq.) and QoE-aware (eq. 9) rewards; (4) an honest characterization of what does and does not hold up on real hardware (Section VII).
- Scope statement: single gNB, 3 UEs (one per slice), rfsim – GNN state representation, multi-agent coordination, and federated learning across gNBs (paper #3's remaining scope) are explicitly future work for the journal-length follow-up, not attempted here.

II. Related Work

TODO(MANOJ): Related work. Suggested content sketch:

- Position against papers #1 (MECON) and #2 (ICRAIE): same admission-control MDP formulation and reward (eq. 2), but simulation-only there vs. live here.
- Position against paper #3 (the IEEE Access survey): this work is a grounded first step toward the survey's live-testbed vision, deliberately scoped down (no GNN/MARL/FL yet – single gNB has no multi-agent coordination problem to solve).
- Brief coverage of O-RAN near-RT RIC / xApp literature, and any other live O-RAN RL testbeds you're aware of (if none exist with a real E2 loop, that itself is worth stating explicitly).

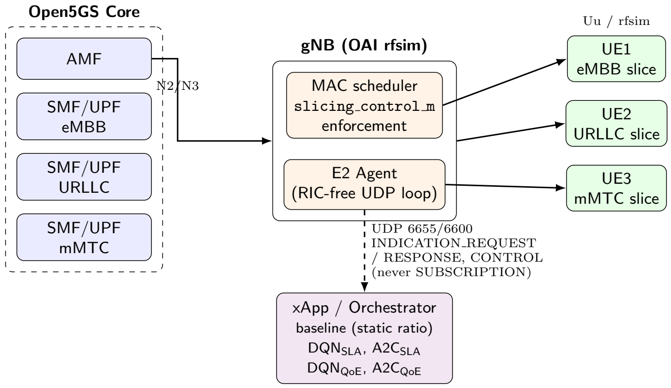

Fig. 1: Testbed control-loop architecture: Open5GS core, OAI gNB (MAC scheduler + E2 Agent), 3 rfsim UEs (one per slice), and the xApp/orchestrator realizing all 5 evaluated arms.

III. Testbed and Implementation

A. Architecture

The testbed comprises a real OpenAirInterface (OAI) gNB running in radio-frequency simulation (rfsim) mode, an Open5GS 5G core provisioned with three slice-specific SMF/UPF pairs, and three OAI `nr-uesoftmodem` UE instances, one per slice (eMBB, URLLC, mMTC), each attached to its own provisioned S-NSSAI. Fig. 1 shows the control-loop architecture. The gNB is built from the ORANSlice fork of OAI (2024.w28 base) [1], which integrates a RIC-free, UDP-based E2 agent directly into the gNB binary – distinct from vanilla upstream OAI, whose only E2 agent (FlexRIC) requires a full E2AP/SCTP near-RT RIC deployment [2], [3].

B. The E2 Control Loop and the Admission-as-Ceiling-Nudging Realization

The gNB's E2 agent exposes a request/response UDP protocol (ports 6655/6600): an `INDICATION_REQUEST` returns one `INDICATION_RESPONSE` carrying per-UE KPM samples (`avg_prbs_dl`, `dl_mac_buffer_occupation`, `dl_bler`, ...); a `CONTROL` message is fire-and-forget. Critically, *the only control primitive this E2 agent exposes is* *`slicing_control_m{sst, sd, min_ratio, max_ratio}`* – a per-slice PRB ratio ceiling/floor enforced by the MAC scheduler [4]. There is no raw per-flow accept/reject hook. Consequently, the admission-control MDP of papers #1/#2 – whose action space is a binary accept/reject decision on an incoming request – is realized here as *ceiling nudging*: an accept decision raises a slice's `max_ratio` toward its calibrated cap, a reject lowers it toward its floor. Capping a slice starves it (demand becomes backlog); it does not remove the demand. This is a structural constraint of any OAI-based live testbed, not an implementation choice, and every result in Section V should be read through this lens.

C. Empirical Ratio-to-PRB Calibration

A first-principles assumption – that the `max_ratio` configuration parameter (0–100 scale) maps linearly onto real PRBs at this cell's 106-PRB/$\mu = 1$ configuration – was tested directly against live measurement and found false: pinning eMBB's `max_ratio` to 4 empirically permitted 15–22 real served PRBs, not 4. Ratio caps for all three slices were therefore calibrated by direct measurement rather than by formula: real per-UE offered demand was polled with the ceiling held wide open (`min_ratio`=0, `max_ratio`=100), and each slice's cap was then set below its own measured organic demand (Table I), reproducing the intended contention regime empirically instead of assuming it. A dedicated contention-gate test confirmed the calibration is load-bearing: pinning eMBB's ceiling to its floor for ∼45 s raised `dl_mac_buffer_occupation` by more than four orders of magnitude ($411 \rightarrow 6.6{\times}10^6$), and restoring a ceiling closer to (but still below) organic demand produced real, measured drainage ($6.6{\times}10^6 \rightarrow 2.5{\times}10^6$) rather than permanent saturation – i.e., ceiling changes have real, reversible, non-trivial consequences on this rig, which is the precondition for any of the policy comparisons in Section V to be meaningful.

D. Baseline Realization

The comparison's static baseline is intended to model the "heuristic, built-in, non-xApp" O-RAN default: a fixed per-slice ratio, no learning, no per-step control. The framework's own `rrmPolicy.json` periodic-reload path, which would provide this natively, was confirmed non-functional on this OAI checkout (the reload code path is compiled out unless a source patch is applied). The baseline is instead realized by pinning each slice's admission-gate ceiling-step size to zero, so no accept/reject decision can move the ceiling away from its calibrated nominal ratio for the entire episode – verified live: the commanded ceiling is bit-for-bit identical on every step of every baseline episode (Fig. **??**).

E. Learned Arms

Two algorithms (DQN, A2C) are each trained offline under two reward formulations: the SLA-only reward of papers #1/#2 (Eq. 2) and a QoE-aware reward $r = \alpha \cdot \text{MOS} - \beta \cdot \text{cost} - \gamma \cdot \text{SLA\_viol}$ (Eq. 9, $\alpha{=}1.0, \beta{=}0.2, \gamma{=}0.5$), where MOS is inferred per slice via an IQX closed-form prior [5] refined by a small per-slice LSTM, calibrated against ITU-T P.1203 [6] objective labels for eMBB and ACR-style task-success scoring for URLLC/mMTC. All four learned arms are trained offline against a closed-loop synthetic KPM source (demand response to admission decisions, not open-loop) for 300 episodes across 3 seeds, then evaluated live with frozen weights – live episodes never update policy parameters.

IV. Experimental Setup

Table I summarizes the fixed testbed and campaign parameters. All five arms (baseline, DQN_{SLA}, $A2C_{SLA}$, DQN_{QoE}, $A2C_{QoE}$) are evaluated against the identical per-slice traffic profiles, episode length, and evaluation-seed set, with arm order interleaved across seeds (not run back-to-back per arm) to decorrelate any single arm's results from time-dependent rig drift. Ceilings are reset to a wide-open state and backlog is drained (verified by re-probing real demand) between every arm.

TODO(MANOJ): Note once Section V is finalized: if the actual completed grid differs from the 3 seeds $\times$ 5 episodes $\times$ 5 arms planned here (e.g., an arm fell short after retries per the campaign's own stop conditions), Table I and this section must be updated to state what was actually run, not what was planned.

V. Results

TODO(MANOJ): RESULTS PLACEHOLDER – to be populated from experiments/RESULTS_REPORT.md once the live campaign (Phase A) and analysis (Phase B) complete. Do not write paper prose here manually; the numbers, tables, and figure narration in this section must be generated/transcribed directly from RESULTS_REPORT.md so every quantitative claim traces to a run ID.

VI. Discussion

TODO(MANOJ): Discussion framing – suggested content sketch:

- What the live results confirm vs. what remains open relative to the simulation-only papers #1/#2.
- The QoE-vs-SLA-reward tradeoff tension, if the URLLC MOS finding replicates at multi-seed scale (Section V/VII will state the verdict; this section is where you interpret it for the reader).
- Positioning for the journal-length follow-up: GNN state representation, multi-agent coordination, and federated learning across gNBs are natural extensions once a multi-gNB physical rig exists – explicitly out of scope here.

VII. Limitations

This work has the following limitations. **Radio interface.** All results are from OAI's rfsim radio-frequency simulator, not an over-the-air deployment; rfsim reproduces the full protocol stack and real scheduler behavior but not physical-layer channel impairments. **Hardware.** The testbed runs on a single shared desktop machine (7.4 GB RAM, 8 cores), not an isolated telecom-grade rig; running the full gNB + 3-UE stack was found to be subject to an intermittent radio-link failure (RLC maximum-retransmission on the highest-traffic slice), plausibly caused by missed real-time scheduling deadlines under memory pressure rather than a logic fault. This was mitigated, not eliminated: a health-check-and-restart mechanism runs automatically between evaluation episodes and was verified live to recover automatically in every observed instance, at a wall-clock cost of roughly one restart per episode transition under full traffic load; every evaluation episode counted in Section V completed and was log-verified end to end, and no partial or corrupted episode was silently included. **Scale.** A single gNB and three UEs (one per slice) were used; the multi-gNB load-balancing term from paper #2 is consequently inapplicable here ($\rho \equiv 1$ with one gNB) and is not evaluated. **Sample size.** Live evaluation used 3 seeds per arm; Section V reports variance honestly rather than treating $n = 3$ as sufficient for strong statistical claims. **QoE labels.** Per-slice MOS is an *inferred* quantity (IQX prior + LSTM refinement calibrated against P.1203/ACR proxies), not a measurement from real end-user subjective scoring.

Reproducibility

ORANSlice commit `b9bcc9b1` (OAI 2024.w28 base; full hash: b9bcc9b17fbecfc1041072a7b8d0f01ae874aba2). Evaluation seeds: 950, 951, 952 (live); offline-training seeds: 256, 257, 258. All configs (`experiments/configs/*.yaml`), orchestration scripts, and raw per-step logs (`omega_log.jsonl`, one row per control-loop step with a non-empty `limitation` field by construction) are preserved and version-controlled; every figure in this paper is regenerated directly from those logs by a committed script in `experiments/plots/`, never hand-plotted.

VIII. Conclusion

TODO(MANOJ): Conclusion – 3-4 sentences summarizing the contribution and the single most important number/finding from Section V, plus a one-sentence pointer to the journal-length follow-up (paper #3's full GNN-MARL-FL scope).

References

[1] WinesLab, "ORANSlice: O-RAN network slicing on OpenAirInterface," https://github.com/wineslab/ORANSlice, oRANSlice fork, OAI 2024.w28 base, commit b9bcc9b. VERIFY: confirm citation format WinesLab/Northeastern prefers for this repo.

[2] OpenAirInterface Software Alliance, "OpenAirInterface: 5g software alliance for democratising wireless innovation," https://openairinterface.org, vERIFY: check for a canonical OAI citation (e.g. a EuCNC/WCNC demo paper) vs. the project URL.

[3] O-RAN Alliance, "O-RAN working group 3, near-RT RIC architecture and E2 general aspects and principles (E2AP)," O-RAN Alliance, Tech. Rep., 2023, vERIFY: exact spec number/version (e.g. O-RAN.WG3.E2AP-v03.00) and year.

[4] ——, "O-RAN working group 3, E2 service model (E2SM) KPM," O-RAN Alliance, Tech. Rep., 2023, vERIFY: exact spec number/version (e.g. O-RAN.WG3.E2SM-KPM-v03.00) and year.

[5] M. Fiedler, T. Hossfeld, and P. Tran-Gia, "The IQX hypothesis: A fundamental property of quality of experience," *IEEE Network*, 2010, vERIFY: exact volume/issue/page numbers.

[6] ITU-T, "P.1203: Parametric bitstream-based quality assessment of progressive download and adaptive audiovisual streaming services over reliable transport," ITU-T, Tech. Rep., 2017, vERIFY: exact edition/amendment referenced (base P.1203 vs. P.1203.1/.2/.3).

TABLE I: Testbed and campaign parameters

Parameter	**Value**
RAN configuration	
gNB	OAI rfsim, ORANSlice fork (OAI 2024.w28 base)
Bandwidth / numerology	106 PRB / $\mu = 1$ (30 kHz SCS), band 78
E2 control primitive	`slicing_control_m`{sst,sd,min_ratio,max_ratio}
E2 transport	UDP 6655/6600, request/response (no SUBSCRIPTION)
Slices (S-NSSAI)	
eMBB	SST=1, SD=0xFFFFFF
URLLC	SST=1, SD=0x000001
mMTC	SST=1, SD=0x000002
Traffic profiles (per slice, fixed, never varied across arms)	
eMBB	sustained UDP, 4 Mbps, 1200 B packets
URLLC	sustained UDP, 300 kbps, 100 B packets
mMTC	bursty UDP, 50 kbps, 80 B packets, 2 s on / 6 s off
Calibrated PRB ratio ceilings (empirically derived, §IV)	
eMBB `max_ratio_cap`	12 (organic demand ≈15 PRB mean)
URLLC `max_ratio_cap`	4 (organic demand ≈5 PRB floor)
mMTC `max_ratio_cap`	3 (organic demand ≈5 PRB floor)
QoE reward (eq. 9) weights	
α (MOS) / β (cost) / γ (SLA viol.)	1.0 / 0.2 / 0.5
Evaluation campaign	
Arms	baseline, $\mathrm{DQN_{SLA}}$, $\mathrm{A2C_{SLA}}$, $\mathrm{DQN_{QoE}}$, $\mathrm{A2C_{QoE}}$
Episode length	60 steps $\times$ 5 s = 5 min
Seeds (paired across arms)	950, 951, 952
Episodes / arm / seed	5
Total live episodes	75
Offline training (per learned arm)	300 episodes $\times$ 3 seeds (256, 257, 258)